

Towards Explainability of non-Convolutional Neural Networks

Jonas Doese
jonas.doese@uni-due.de
PhD Student
University Duisburg-Essen
Duisburg, Germany

Torben Weis
torben.weis@uni-due.de
PhD Supervisor
University Duisburg-Essen
Duisburg, Germany

ABSTRACT

Artificial intelligence has risen in popularity in research and applications in the past years. Explainability is a topic that has been proven to work on human interpretable data like images or sentences, but the research is narrow whenever such data is missing.

Creating trust through explainability on raw data processing neural networks of any kind will become necessary in the future as networks are evolving further towards artificial general intelligence.

This research is focused to visualize parts of the hidden layers instead of focusing explainability on the input data and is independent of the neural network's size. We create a model that represents the neural network in a way that neurons that are activating on similar features are grouped together in structures. This model will be analyzed in a machine-learning equivalent process to identify parts of the network being responsible for a decision.

In a further step we use the model to test the processing of raw sensor-data versus an approach of heatmapped explainability with a convolutional neural network. Relevant data points in the input are visualized by a common heatmap approach while the hidden layers are analyzed in this research and should point to structures that have a comparable function in the network. For example, if the heatmap highlights peaks of values, the model will be highlighted in the area that is observed as the activation of the neural network on peaks. We will provide the research of artificial general intelligence with a solution for explainability which is necessary for advanced research and the operation of such applications in complex or dangerous scenarios.

CCS CONCEPTS

• **Artificial intelligence** → **Explainability**; *Artificial general intelligence*; • **Neural networks** → Recurrent neural networks; Analysis.

KEYWORDS

artificial intelligence, artificial general intelligence, neural networks, CNN, RNN, explainability, interpretability, trust

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1 INTRODUCTION

While current approaches in explainability are focused on analyzing the input vector to get human-readable results, our approach will be able to generate a model for visualization of the hidden layers.

This way we provide a solution that is independent of the neural network's size and are able to give explainability on huge counts of neurons.

First the relation between artificial neurons is described as a distance value inspired by axons, which are the connections between neurons in the brain. With this distances neurons are grouped together and a representation of the neural network is build that will show regions like a brain in a MRI scan. To reduce the complexity of the visualization model, closely related neurons are collapsed iteratively.

Given a certain input vector, areas in the model are colored based on their activation which makes it possible to compare result to other input data.

Getting further explanations on the input, the areas in model will be labeled with the features they activate on. This requires more analysis on the model with feature-masked data and an approach similar to training a neural network.

In the end an evaluation will be done on time-series sensor data, that will be computed with different types of networks. Using a convolutional neural network, common approaches of heatmapping will show differences for the explainability with our work. Other types of neural networks will show if explainability can yield a similar quality regardless of the network's design.

2 RELATED WORK

Current research shows the increased use of artificial intelligence in various scenarios [3] and an extended use in applications. A lot of different types of neural networks have been developed to be used for certain applications.

Explainability primarily focuses on the ability to analyze a trained neural network by observing it's behaviour. Further this can be extended on the fields of trust and interpretability.

Trust is a human-user-centralized approach to give a person insights about a decision process of an artificial intelligence while this person is not familiar with neural networks. Explainability

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can be congruent with trust as a visualized result can possibly be designed to help inexperienced users.

Unsupervised trained neural networks and any solution vector that can not be matched intrinsically on human-understandable information has to be processed so that a user is able to understand it. Algorithms that cover this applications are grouped into interpretability [2, 4]. Expanding the output vector of a neural network in a way to get more information than actually needed and processing it further in terms of interpretability can yield explainability.

Research has been done over the last years to improve explainability on convolutional neural networks by highlighting features on the input image. The approach to overlay heatmaps on input data images has been picked up multiple times [7, 11, 12] and different algorithms have been described to improve the heatmap's quality. The heatmap is displayed as colored pixels on the original input, where deep red shows areas of interest to the neural network. Showing the interest of explainability as a feature for artificial intelligence in general, the approach using heatmaps is adopted onto recurrent neural networks as well [7].

Yet a single paper exists about explainability of fuzzy logic in neural networks [10], describing a general approach towards explainability of black boxes in neural networks.

Common approaches for explainability of neural networks are described in various metrics to evaluate a given solution from the output to the input layer through the network. These metrics were tested with convolutional neural networks and displayed with an overlay heatmap. Explainability was generated by the heatmapped features in the image which is possible because all input neurons can be traced back onto the image. As every pixel is connected to an input neuron, the relevance of input neurons can be visualized on the pixels.

Another approach focused on feature extraction of images similar to the grouping of areas in the neural network we use, but on areas in the image input [6]. Together with the solution of the artificial intelligence that was used to identify birds, explanations were given why the bird was classified this way.

This helped to prove that a convolutional neural network that was trained to identify vehicles, detected ships by the water surrounding it [13]. This proof only considered inputs and outputs while not looking inside the network.

By using a k-Nearest Neighbor algorithm it is possible to analyze hidden layers to deduct biases of the learning process [9].

Only a few approaches exist that try to build a comparable level of explainability, interpretability and trust in recurrent neural networks. Even fewer work exists on explaining the network itself independently of its design.

Artificial general intelligence focuses research on extending neural networks to broader applications and not specific problems. This leads to more complex networks with input vectors far bigger than a single image or a sentence on which current explainability approaches are not possible [1, 5].

Similar to our approach, Kazhdan describes a model extraction on recurrent neural networks with domain-knowledge to classify activations in the network with human concepts [8].

3 METHODOLOGY

Artificial neurons lack the information of how close they are related to each other. Neurons in the brain are connected by axons, which inspired this research to add distance values to the artificial neuron connections. These distances resemble how much of an activation in one neuron is affected by that of another neuron.

Distance values also bring more information than the already used weights on connections, as weights are only useful in the context of all weights on a neuron and the bias. This means that a high positive weight can also be redundant if its effect is absorbed by a high negative bias and vice versa.

We add a distance value to every connection between neurons. This distance value will be used to perform a grouping of neurons in structures of similar functionality. A structure will contain all neurons that have a distance below a certain threshold from one starting neuron which is chosen randomly. Repeating this process on every unstructured neuron is described in the section Structuring.

For huge neural networks we will reduce the data points in the model by collapsing neurons into single representations called clusters. This will help rendering the structures and reduces computation time in a live view of the artificial intelligence. Algorithms and problems on clustering are described in the section Clustering.

The final part can be done on structures or clusters in the same way, so that clustering is optional but helpful for explainability. Structures and clusters have a lack of information and will not be distinguishable from each other. Explainability for convolutional neural networks with heatmapped images benefits from the image that the user can observe. If for example an ear in an image was the reason for the neural network to determine the content of the image as a specific animal, this ear will be colored by the heatmap. Coloring clusters won't lead to the same level of explainability because there is no real-world connection to what this cluster responsible for. Using approaches similar to machine-learning, we will give clusters an explanation necessary for explainability as shown in section 3.3.

3.1 Structuring

We use the following two metrics as a possibility to generate structures within a neural network. As stated before, the weight multiplicant that is already used in connections between neurons is not a suitable metric for this task. This is because weights can assume negative and positive numbers which in combination can make a single high value not representative in the sum of weights. Additionally, weights are relevant in combination with other weights in a layer and in combination with a bias.

We cannot use weights directly, but it is possible to sum up all weights of a single neuron and compute the relative impact of individual weights to normalize the values. With this normalization we calculate the distances for each neuron and are able to compare them. By grouping neurons together based on their distance the network will adopt to a structured solution. First the structures will be anonymous as they are grouped together, but give little insight about *why* they are structured. Through domain knowledge we then assign labels to these structures and perform a classification.

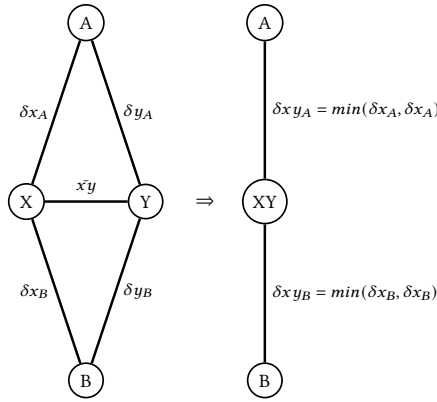


Figure 1: Collapsing two neurons with minimum-metric

Alternatively it is possible to use machine learning in a similar way to adopt structures with masked input data. This approach has been used on images to achieve explainability with feature classification [6].

A masked feature is fed into the neural network and as the activation of neurons can be measured, it is possible to group neurons that activate on similar features by moving them towards each other in a model representation. With more and more data the neurons will approach a structured form of similar functionalities which inherits distance values.

This will lead to inherit distance values as structures will emerge in the learning process. If the distance values are needed for proceeded clustering they can be computed from the model. To reduce the number of steps towards the final distance values, the first metric can be used as an initial state.

In either case, with no knowledge over the input data it is impossible to link structures to real-world properties and therefore it is impossible to enable explainability.

3.2 Clustering

Clustering the structured neural network can be used to speed up live rendered activations. While a most neurons will be grouped close together, it is not necessary to keep them all in the visualization. The algorithm is based on the computation of Level-Of-Detail models in computer graphics where vertices are combined to reduce the polygon count of the model. Close neurons will be collapsed into a single neuron while keeping the minimal distance to all connected neurons as shown in Figure 1. This approach will be executed iteratively until the plot is at an optimal state for visualization. This state can be found by comparing iterations and choosing one that fits the needs for further explainability research. Other metrics on collapsed distances have to be evaluated.

Figure 1 shows a connection of four neurons, where the neurons X and Y will be replaced by a single neuron by keeping the minimum distance as the new distance for the combined neuron. The problem of collapsing neurons like in Figure 1 occurs whenever meshes of three neurons are possible, which is not the case in feed forward networks. Even if the problem does not occur in the first place, it can happen that the model of the network forms such connections

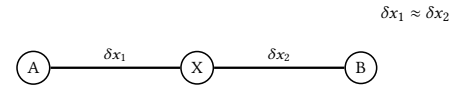


Figure 2: Structuring problems

of three neurons after some iterations and neurons are collapsed. Another problem could be that a single neuron has approximately equal distances (Figure 2) to two or more other neurons that are located in different structures. In this case there are three options. The neuron can be ignored and left in place, such that it will not be collapsed to either structure. Another option is to collapse the neuron randomly or based on a metric towards either structure. The last option is to show the proximity to both structures by cloning the neuron and collapse one of them to each structure. This way a visualized activation of that neuron can be observed in both structures.

3.3 Describing

Structures and Clusters lack the information of what their activation means to the output of the neural network. Therefore, domain-knowledge is needed to finally describe and label structures in a way that a human can understand what happens with an activation. Without describing the visualization it is only possible to check if a given input activates similar or different structures compared to another input. By generating labels with masked input values, structures can be linked with real-world properties, making it possible to tell what the activation of a structure has caused. The computed distances in either way can be plotted as a plot of the neural network. This visualization can be used when analyzing the computation on unknown data sets whenever the input data itself is not helpful to experience explainability. By highlighting activating structures on a given input the neural network can be observed.

4 EVALUATION APPORACH

As shown in the TED-talk by Blaise Agüera y Arcas [13] it is possible to revert a network to visualize its "creativity". The results show that by processing a neural network in reverse, from an output to the input, the networks "creativity" emerges and can be visualized [13]. In an experiment, a convolutional neural network for face recognition was tasked to draw the perfectly detectable human face and it emitted an image containing multiple noses and eyes. The explanation is that the network is mostly activating on nose and eye structures.

With a perfectly clustered network, it should be possible to task the same network to draw the perfect eye, making it possible to only analyze parts of a given network.

The described metrics can be compared visually and optimized further towards a better clustering.

In a first attempt image based convolutional neural networks will be used with state-of-the-art explainability approaches, most notably heatmapping on the input. By using our explainability approach alongside the known approaches, similarities in the approaches will show up, giving another possibility to optimize the clustering. With a working metric and clustering a non-convolutional neural

network can be described. In our case the goal is to analyze and describe recurrent neural networks working on time based sensor data. In the end it should be possible to state what happened based on the activations in the network and giving the explainability on a level that is understandable by any human.

A trained convolutional neural network can be described by the visualization and compared with heatmap approaches. It should be possible to tell either way if a certain feature or the absence of that feature triggers a group of neurons towards a decision.

5 EXPECTED CONTRIBUTION

Convolutional neural networks working on imagery data can yield explainability through the input data. These approaches don't work on context-less input data as there is no visualization for huge input vectors that are human readable.

By clustering and structuring neural networks in general it should be possible to analyze any artificial intelligence independently of the underlying design pattern. We also look further into the black box of a neural network, instead of just mapping input and output together. This allows to achieve explainability on a different level that also changes how a user experiences it. With the given domain-knowledge users can identify problems in neural networks, biases and less trained features.

Possible use cases could be to detect biases in neural networks and for example retrain this part instead of recreating the artificial intelligence.

In general the analysis of the black boxed structures in neural networks helps with the emerging necessity in transparency and ethics of evolving applications with artificial intelligence in our life. This also allows more complex neural networks to be developed for problems that are more than just images.

The bigger the input vector is, the more data is processed in an artificial intelligence. In the terms of multiple sensors and cameras like in today's cars or production machines, an accepted solution has to consider all this data as relevant. Our approach will provide the scalability on current research problems in artificial general intelligence and an improvement in the development of complex neural networks in various projects.

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BIBLIOGRAPHICAL SKETCH

Jonas Doese is a PhD student and research assistant at the chair of Distributed Systems at the University of Duisburg-Essen. He is supervised by Professor Torben Weis, who leads the chair of Distributed Systems as part of the Faculty of Engineering. His research focuses on human-centered artificial intelligence in the terms of explainability, interpretability and trust in various applications. He started his PhD in May 2021 and is expected to complete in 2024.